BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Defensive Design

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does authentication establish?

2. Why translate high-level source code?

A) The CPU only reads comments
B) It increases physical RAM
C) The CPU needs machine code
D) It removes every logic error

3. "Sequence means the computer chooses any random instruction order." Correct this. Explain why.

4. Insertion sort has sorted part [2, 5, 8] and next value 9. Which values move? Give the new sorted part.

Today's objectives

Explain the core ideas of defensive design using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Anticipate misuse

Reject unexpected input and avoid exposing sensitive detail in errors.

Input sanitisation

Handle or remove harmful/unexpected characters where appropriate.

Maintainability

Meaningful identifiers, comments and modularity make software easier to inspect.

Defensive design

Plans for invalid data, malicious use and foreseeable user mistakes.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Anticipate misuse	Reject unexpected input and avoid exposing sensitive detail in errors.
Input sanitisation	Handle or remove harmful/unexpected characters where appropriate.
Maintainability	Meaningful identifiers, comments and modularity make software easier to inspect.
Defensive design	Plans for invalid data, malicious use and foreseeable user mistakes.

I Do - One worked example

Improve a login counter that accepts unlimited guesses and prints the correct password on failure.

1. Remove any display of the secret.
2. Limit repeated unsuccessful attempts.
3. Present a non-revealing error and lock/slow further attempts.
4. Keep attempts in an appropriately scoped variable; test boundary conditions.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A text form permits unescaped SQL input. Suggest improvements.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the condition that locks an account after three failed attempts.

```
attempts = 3
if ... then
print("Locked")
else
print("Try again")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. State two defensive programming techniques. [2 marks]

Q3. Explain why meaningful variable names aid maintainability. [2 marks]

Q4. Explain why showing a user's actual password in an error message is unsafe. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Input validation is one defensive design technique. ☐ **T** ☐ **F**
2. Reject unexpected input and avoid exposing sensitive detail in errors. ☐ **T** ☐ **F**
3. Handle or remove harmful/unexpected characters where appropriate. ☐ **T** ☐ **F**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Whether a user has supplied suitable evidence of the claimed identity. ☐

Q2 C - The CPU needs machine code ☐

Q3 Instructions execute in the specified order. ☐

Q4 (a) Values greater than 9 shift right: []. New prefix [2, 5, 8, 9]. ☐

Code First - Model

```
attempts = 3
if attempts >= 3 then
print("Locked")
else
print("Try again")
endif
```

Output: Locked

We Do - Check the model

1. Reject unexpected inputs without leaking data. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Input validation and restricted access or appropriate sanitisation; any two justified techniques. ☐

Q3 They make purpose easier to understand and reduce debugging/maintenance effort. ☐

Q4 It exposes a secret to observers and can enable unauthorised access. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: